

## Multiple Choice (10 marks)

1. Which foods tend to be consumed in lower quantities in Wales and Scotland (as of 2020)?
  - (a) Meat
  - (b) Grains
  - (c) Fruits and vegetables
  - (d) Legumes
  - (e) Seafood
2. Bronchial breathing is characterized by
  - (a) Inspiratory component louder and longer with a gap between inspiration and expiration
  - (b) Inspiratory component louder and longer, but no gap between expiration and inspiration
  - (c) Expiratory component louder and longer, but no gap between expiration and inspiration
  - (d) Both inspiratory and expiratory components are equal in length and volume
  - (e) Inspiratory component louder and longer with a gap between expiration and inspiration
3. How many chromosomes do all human somatic cells contain?
  - (a) 30
  - (b) 24
  - (c) 44
  - (d) 3
  - (e) 23
4. Tay-Sachs disease is caused by deficiency of \_\_\_\_\_?
  - (a) Alpha-L-iduronidase
  - (b) Arylsulfatase A
  - (c) Hexosaminidase B
  - (d) Glucose-6-phosphatase
  - (e) Lysosomal alpha-glucosidase

5. A difference between the social networks of older and younger adults is that older adults have
- (a) Larger networks overall
  - (b) Less frequent communication with their circles
  - (c) Smaller networks overall
  - (d) More dispersed circles
  - (e) Smaller inner circles

## **True / False (10 marks)**

*Answer True or False*

- 6. True or False: Cohort effects primarily impact chronic conditions like diabetes rather than behavioral issues such as alcoholism.
- 7. True or False: Identity is primarily determined by one's financial status and societal position within a community.
- 8. True or False: The intercostal nerve lies medial to the inferior border of the corresponding rib.
- 9. True or False: Infants and children require the most protein-dense foods to support their rapid growth and development.
- 10. True or False: The pituitary gland is located within the ventral body cavity, specifically in the cranial cavity.

## **Long Form Response (20 marks)**

*Answer each question with a clear and complete explanation.*

- 11. Discuss the implications of altered breathing patterns in patients with head injuries and altered consciousness, and explain why anaerobic respiration is not classified as one of these patterns.
- 12. Discuss the role of vitamin D as a major lipid-soluble antioxidant in cell membranes, including its mechanisms of action and implications for human health.

*End of Paper*